

Graph Theory

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1 Introduction

1. Let G be connected and $e \in E(G)$ an edge. Then $G - e$ has at most two components.

The idea is to partition $V(G)$ into vertices that are connected by a path not containing e to an endpoint of e . Let $e = \{u, v\}$, $G(u) = \{x \in V : P(x, u) \in \{\mathbb{P}(x, u) - e\}\}$ and define $G(v)$ similarly. Since G is connected, let $P(x, u) = \{e_1, e_2, \dots, e_k\}$ be a path with $e_i = \{x_{i-1}, x_i\}$, $x_0 = x$, $x_k = u$. If $e \notin P$, then we're done. Else, let $j \in [k]$ be minimal such that $e_j = e$. If $e_{j-1} = u$, then $Q(x, u) = \{e_1, e_2, \dots, e_{j-1}\} \in \{\mathbb{P}(x, u) - e\}$. Else, $Q(x, v) \in \{\mathbb{P}(x, v) - e\}$, so $x \in G(u)$ or $x \in G(v)$.

Note $G(u), G(v)$ are connected in $G - e$, so either these are the components we want or $G - e$ is connected.

Note if $P(x, y) \in \mathbb{P} - e$ for $x \in G(u), y \in G(v)$, then $P(u, x), P(x, y), P(y, v) \in \mathbb{P} - e$, so $P(u, v) \in \mathbb{P} - e$.

If $P(u, v) \in \mathbb{P}$, then there exists $Q(u, v) \in \mathbb{SP}$. Let $P = \{e_1, e_2, e_3, \dots, e_k\}$, $e_i = \{x_{i-1}, x_i\}$, for $x_0 = u$, $x_k = v$. Let a_i be a subsequence of $0 \cup [k]$ with $a_0 = 0$. Given $a_i \neq k$, pick j maximal such that $x_{a_i} = x_j$. If $j = k$, we're done. Else, let $a_{i+1} = j + 1$, so $x_{a_{i+1}} \neq x_{a_0}, x_{a_1}, \dots, x_{a_i}$. The path $Q(u, v) = \{e'_i\}$ for $e'_i = \{x_{a_{i-1}}, x_{a_i}\}$ is simple.